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Design and Analysis of a D-Band (110-170 GHz) Active Balun with Low Amplitude and Phase Imbalance in 130nm SiGe BiCMOS Technology

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Outline

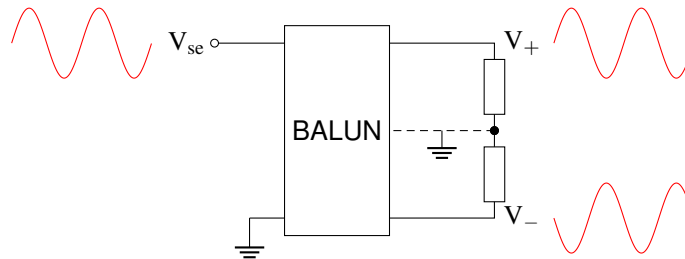
- Introduction
- State of the art
- Technology
- Circuit design
- Layout
- Results
- Comparison with the state of the art
- Conclusions



Introduction

Objective of the work

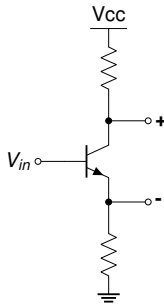
BALanced - UNbalanced



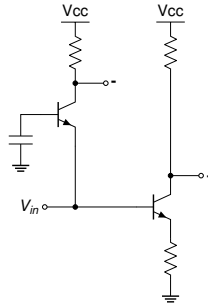
- Design and analysis of an active balun for D-band applications.
- Main objectives:
 - low amplitude and phase imbalance from 100 to 200 GHz;
 - good efficiency and high gain.

State of the Art

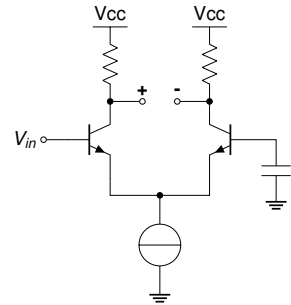
Main active balun topologies



(a) Common-Emitter



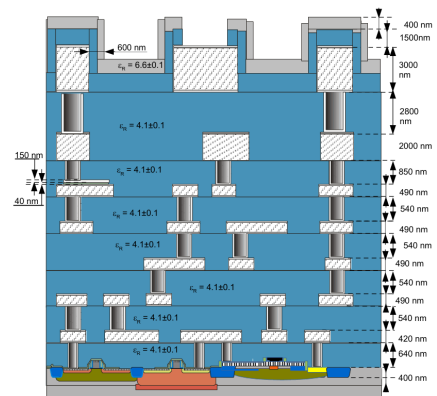
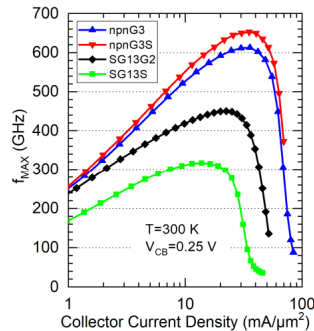
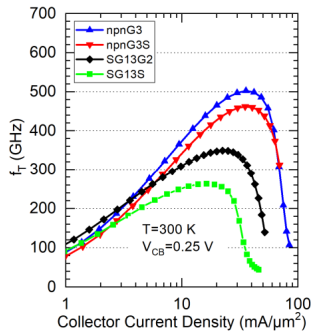
(b) CB + CE



(c) Differential pair

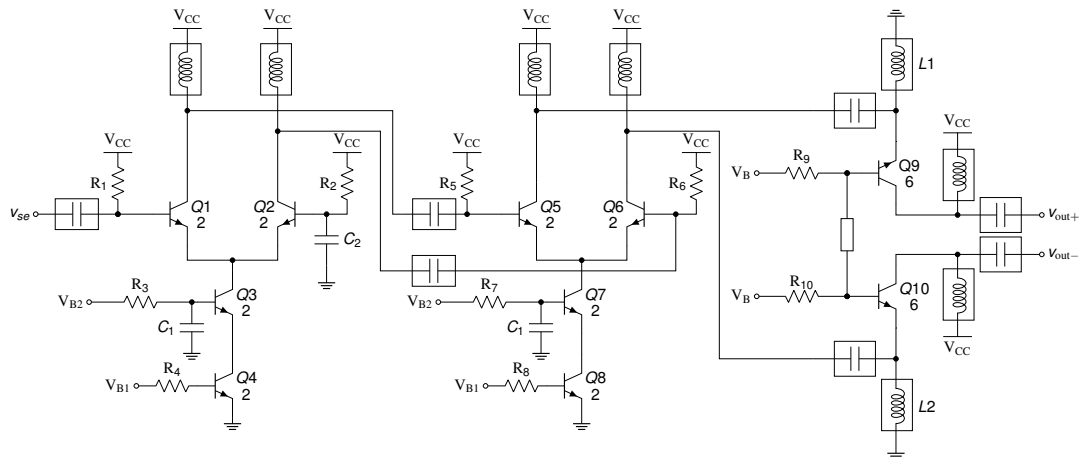
Technology

SG13G3 SiGe BiCMOS



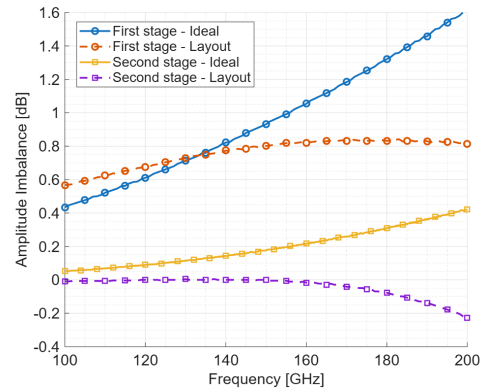
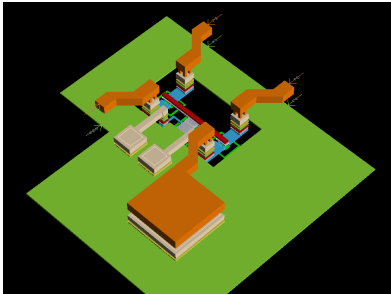
Circuit Design

Schematic-level implementation



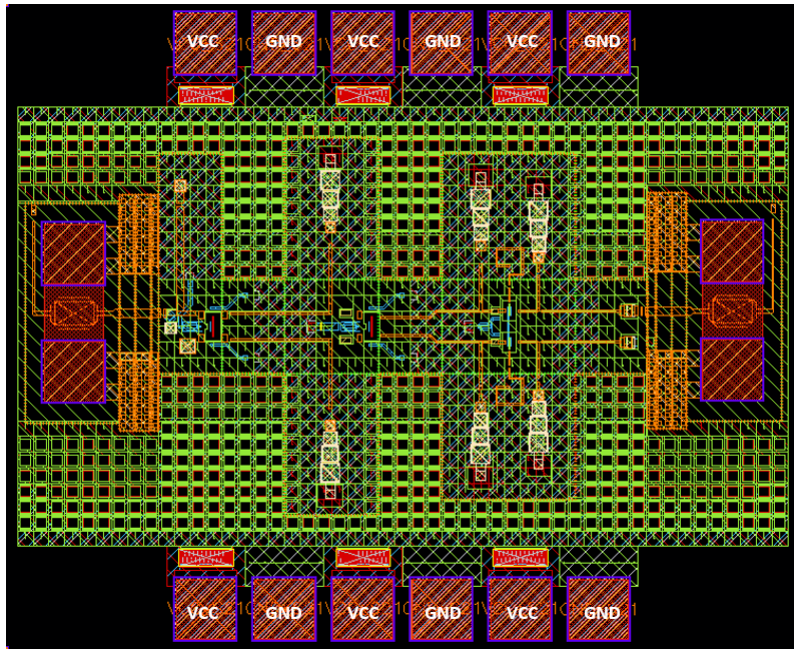
Layout

Layout overview and intermediate results



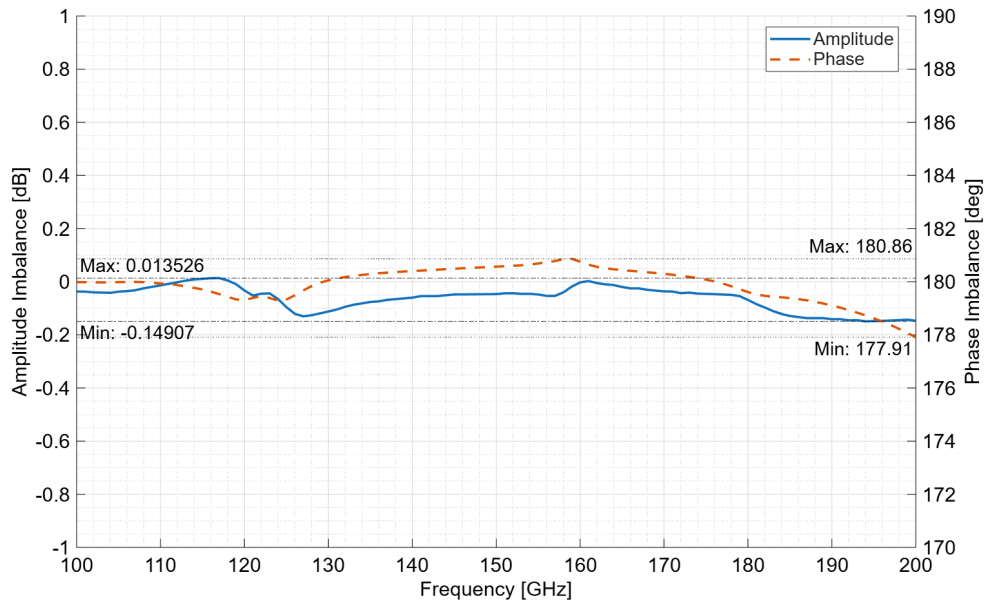
Layout

Final layout implementation



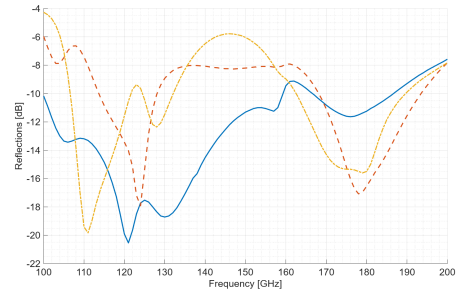
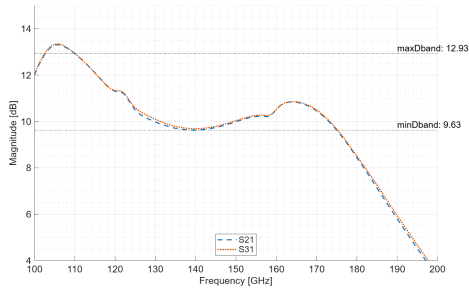
Results

Amplitude and phase imbalances



Results

S-parameters



Comparison with the State of the Art

Reference	[1]	[2]	[3]	This work
Type	Active			
Technology	SiGe	SiGe	SiGe	SiGe
Supply (V)	3	4	3.3	1.8
Frequency (GHz)	237–277	DC–(>70)	DC–70	110–170
Amplitude imbalance (dB)	0.5	0.2	0.65	0.15
Phase imbalance (deg)	5	5	3.2	± 1.5
DC Power (mW)	59.7	144	37	53
Area (mm ²)	0.0168 ^a	0.42	0.42 ^b	0.347 ^c

^a Active core area, ^b Pads included, ^c Pads excluded

[1] J. Hebel, A. Ç. Ulusoy and T. Zwick, "Active BALUN with 40 GHz bandwidth at 257 GHz in 130 nm SiGe:C," 2022 IEEE 22nd Topical Meeting on Silicon Monolithic Integrated Circuits in RF Systems (SiRF), Las Vegas, NV, USA, 2022, pp. 9–12, doi: 10.1109/SiRF53094.2022.9720063.

[2] P. Stärke, V. RieSS, C. Carta and F. Ellinger, "Active Single-Ended to Differential Converter (Balun) for DC up to 70 GHz in 130 nm SiGe," 2019 IEEE BiCMOS and Compound Semiconductor Integrated Circuits and Technology Symposium (BCICTS), Nashville, TN, USA, 2019, pp. 1–4, doi: 10.1109/BCICTS45179.2019.8972713.

[3] A. Franzese, M. H. Eissa, T. Mausolf, D. Kissinger, R. Negra and A. Malignaggi, "Ultra Broadband Low-Power 70 GHz Active Balun in 130-nm SiGe BiCMOS," 2020 IEEE BiCMOS and Compound Semiconductor Integrated Circuits and Technology Symposium (BCICTS), Monterey, CA, USA, 2020, pp. 1–4, doi: 10.1109/BCICTS48439.2020.9392960.

Conclusions

Final remarks

- The proposed balun shows good performance in the 110–170 GHz band.
- The amplitude and phase imbalances remain limited over the frequency range of interest.
- Possible future developments include full post-layout verification and experimental validation.



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Thank you for your attention